

1. What is drawing? Types of Drawing? Why drawing is important for engineer?

A drawing is a graphical representation of an idea, a concept or an entity which actually exist in life.

Types of drawings:

I. Artistic drawing: Artistic drawings range in scope from the most simple line drawings to the most famous paintings. This type of drawings is used to express feelings, belief, philosophies and idea of an artist.

II. Technical drawing: A technical drawing is a means of clearly and consciously communicating all of the information necessary to transform an idea or a concept into reality. It has two types:

(a) Preliminary drawing

(b) Working drawing

Drawing is important for engineers. Some of the reasons are given below:

- ✓ Drawing is the language of engineers. An engineer must be well conversant with drawings.
- ✓ Drawing represent the reduce shape of the structures and owner visualizes what is going to will happen.
- ✓ Drawing provides instructions and guidance to architects, engineers and workmen at field to construct the structure according to the dimension shown in the drawings.

2. Write down the names & functions of drawing instruments.

1. Drawing Sheet (560mm x 355mm)
2. Scotch Tape
3. Drawing Board with parallel bar - Drawing board is used for placing the drawing paper on it with the help of either cellophane tape or board pin.
4. T-square (Tee Scale) - T-square is used to draw horizontal line.
5. Set-square - They are used together with the straightedge of the T-square to draw the vertical and the inclined lines.
6. Triangular scale or Flat scale
7. Drawing pencils(2H, HB, 2B)
8. Pencil sharpener
9. Pencil eraser (Tee Scale)
10. Pencil pointer or Sand paper
11. Instrument box
12. Lettering template
13. Dust cloth (Clean handkerchief)

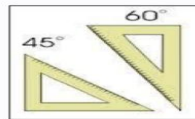
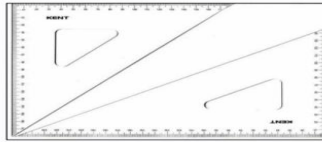
Drawing instruments

- **T-Square (T scale):** T-Square is used to draw horizontal line. It is used on a drawing board. It must be made of transparent plastic. The head of the T-square is placed on the left edge of the board for the right handed person and vice versa.



Drawing instruments

- **Triangles / Set squares:** Triangles are set of 45° and $30^{\circ} - 60^{\circ}$ triangles which are made of clear plastic. They are used together with the straightedge of the T-square to draw vertical and inclined lines.



Drawing instruments

- **Scales:** Scales are used for measurement only. not for drawing lines.
- **Compasses:** Compasses are used to draw a circle or an arc. Most common type compasses are Friction head compass and Drop bow compass.



Drawing instruments

- **Protractor:** Angles can be measured with this tool.



- **Paper:** Drawing paper / Art paper are used for drafting.
- **Pencils and leads:** HB and 2B/3B pencils are used to draw. 0.5mm lead can be used also.
- **Handkerchief:** to clean the paper.

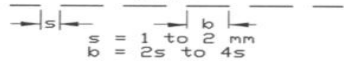
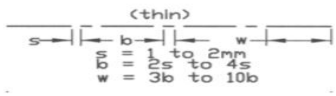
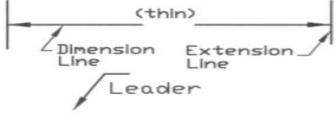
3. What is the scale of drawing? Why is it important?

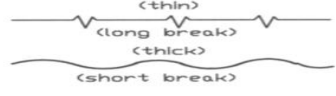
Scale is the ratio of the linear dimension of an element of a machine part as represented in the original drawing to the real linear dimension of the same element of the machine part itself. The scales of drawings are of three categories.

(1) Full Size (2) Enlargement Scale (3) Reduction Scale

4. Briefly explain the types of lines with their usages.

Table 2.1: Types of Lines with Their Usage (contd.)

Sl No.	Types of Line	Usage
1	Visible line / object line 	to indicate all visible outlines of an object. It shows the shape of an object.
2	Hidden line / dashed line 	to represent the hidden edge of an object. It must begin and end with a dash touching the visible lines. Dashes that show hidden lines usually touch each other at intersection.
3	Center line 	to show the center line of holes, pitch line.
4	Extension line, dimension line and Leaders 	to show dimension of an object extension line, dimension line and leaders are used.
5	Section line 	to indicate the cut portion of an object.
6	Cutting plane line 	to show the imaginary cutting of an object

Sl No.	Types of Line	Usage
7	ISO Cutting Plane Line 	
8	Break line 	to show a break on the object. It shortens the view of a long part.
9	Phantom line/repeat line 	to show the alternate position of an object or the position of an adjacent part.

5. Write down the rules that are to be remembered while inserting dimensions on a part:

(i)

Extension Line (ii) Dimension Line (iii) Leaders, (iv) Dimensioning of angles,

(v) Dimensioning of single diameter

Dimension Line

- (1) Dimension line should be approximately 10 mm away from the visible line. The spacing between the consecutive parallel dimension lines may also be considered as 10 mm.
- (2) Dimension lines are broken near the middle to allow space for dimension.
- (3) Dimension lines should usually be placed outside the view unless it becomes necessary.
- (4) When the space in between the extension lines is too small to insert dimension line completed with an arrowhead, it may be provided outside the extension line.
- (5) To accomplish it the shortest dimension line should be placed nearest to the outline of the part.
- (6) Centerlines should never be used as dimension lines.
- (7) In case of extra long dimension line or crowding of dimensions, dimension lines inside the view may be permissible.
- (8) When several dimension lines are required side by side along a line, it is recommended to stagger them.
- (9) The dimension line for a part, which cannot be presented completely on the drawings due to its large distance, the free end is terminated in a double arrowhead pointing in the direction in which it could be completed.
- (10) Dimensioning should be given between the views in general.
- (11) Dimensioning should not be duplicated in other views.
- (12) Dimensioning should be made in such a way so that no subtraction or addition is required to define or locate a feature.
- (13) Dimensioning may be inserted on relatively larger view in order to make it clear.
- (14) One system of dimensions either unidirectional or aligned has to be used throughout the drawing.
- (15) Dimensioning to hidden lines should be avoided in general. To accomplish that a sectional view or broken-out section may be used to place dimensions.
- (16) Dimensioning should be made on the view, which represents the shape of the part best.

Leaders

- (1) A leader should always be inclined at an angle of 60° preferably and 45° occasionally (not horizontal or vertical) with a 3 mm horizontal bar.
- (2) A leader is either terminated by an arrowhead on a line or a small dot of about 1.5 mm diameter within the outline of the part.
- (3) Leaders should not be drawn bent except special circumstances.
- (4) Leaders should not cross each other; however, they may be drawn parallel to each other.
- (5) To direct a circle or an arc the leader should be so drawn, if it is imagined to extend it must pass through the center of the circle or the arc.
- (6) All notes and dimensions in a leader have to be provided in the horizontal direction.

Extension Line

- (1) A gap of 1 mm has to be kept in between the extension line and the visible line.
- (2) An extension line should extend about 3 mm from the outermost dimension line.
- (3) Extension lines may cross each other without a break. They may also cross the visible lines without a break. If the extension lines cross arrowheads or dimension lines close to arrowheads, a break may be permissible.
- (4) Centerlines may be used as extension lines. However, it should not be broken while passing the circle.
- (5) Extension lines are usually drawn perpendicular to dimension lines, where there is overcrowding extension lines may be drawn at an oblique angle.

6. Define: (i) Orthographic Projection (ii) projectors (iii) Orthographic view

When the rays of sight are made in a perpendicular direction to the plane of projection, it is called orthographic projection. The word orthographic is obtained from the Greek words: orthos, meaning straight, correct, at right angles to; and graphikus, meaning to write or describe by drawing lines.

In the projection there is a relationship of the different views. It is usual practice to draw the front view first, then the top and side views are drawn in reference to the front view with the help of the vertical and horizontal projection lines.

7. What is Third Angle Projection & First Angle Projection?

Third angle projection: The projection plane upon which the front view is projected is called the frontal plane (a-b-c-d is the frontal plane) and that upon which the top view is projected is called the horizontal plane (d-c-g-h is the horizontal plane) as shown in the figure. While the projection plane upon which the side view is projected is called the profile plane (a-d-h-e is the profile plane). First angle projection: In this figure front view is shown in the frontal plane (e-f-g-h is the frontal plane), top view in the horizontal plane (a-b-f-e is the horizontal plane) and side view on the profile plane (a-d-h-e is the profile plane). Views on the sides of the unfolding box in the First angle projection are given in Figure 3.11. On the other hand the relative positions of the views on the sides of the unfolding box have been shown in Figure 3.12.

8. What is auxiliary view? Why is it important?

In orthographic projection sometimes it happens that some surfaces of the machine parts are not parallel to the plane of projection i.e. they are at an angle. These surfaces are called inclined surfaces. The regular orthographic view of the inclined surface does not give the exact shape; it gives the distorted shape of the surface. Thus the clear feature of the inclined surface of the object cannot be obtained from the regular view. As a result the main objective of the drawing is lost. In order to overcome this problem an auxiliary view is necessary. A picture plane is assumed parallel to the inclined surface called auxiliary projection plane and orthographic projection on that plane is made. It gives the true shape of the inclined surface and preserves the objective of the drawing. Sometimes an auxiliary view replaces one of the regular views.

9. What is sectional view? Why is it important?

A sectional view is that view, which is seen beyond the imaginary cutting plane through an object at right angle to the direction of sight.

Lots of hidden lines are required to show the interior detail of an object. It becomes very difficult to visualize the object thereby jeopardizing the primary objective of the drawing. It represents the interior construction or details of hidden features clearly to the users. The exposed or cut surface is identified with the help of section lining or cross-hatching. Hidden lines and details behind the cutting-plane line are usually omitted unless they are essential for clarity or dimensioning. A sectional view frequently replaces one of the regular views. A regular front view may be replaced by a front sectional view.

10. Draw the following Symbols for Section Lining: cast iron, steel, fiber, insulation, concrete, bronze.

Section Lining



Cast Iron



Sound Insulation



Earth



White Metal,
Zinc, Lead,
Babbitt and
Alloys



Concrete



Water &
Other Liquids



Steel



Cork, Felt,
Leather & Fiber



Rock



Magnesium,
Aluminum



Marble, Slate,
Glass, Porcelain



Electric
Windings,
Electromagnets etc.



Bronze, Brass,
Copper and
Compositions



Fire Brick
and Refractory
Materials



Sand



Rubber, Plastic,
Electrical Insulation



Thermal
Insulation



Wood

11. What is AutoCAD? Why is it important?

AutoCAD is a commercial computer-aided design (CAD) and drafting software application. AutoCAD is the original CAD software used by millions around the world. It can be used to create precise 2D and 3D drawings and models, as well as electrical diagrams, construction drawings, and more.

1.AutoCAD as an architectural planning tool: It comes with a user-interface with built-in design lay-outs. The lay-outs contain numerous templates specially designed for architectural planning and building construction.

2.AutoCAD as an engineering drafting tool: You can use AutoCAD to draw accurate 2D drawings for any engineering domain and also use AutoCAD to render to 3D models to help in visualization of the end product.

3.AutoCAD as a graphic design tool: AutoCAD has innate features that enable users to plan and map out spaces and take advantage of the space available.

4.AutoCAD in 3D Printing.

5.AutoCAD as industrial design tool.